



UAV READY

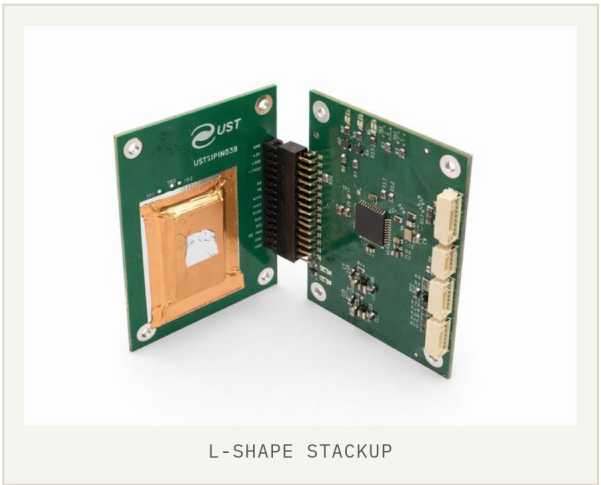
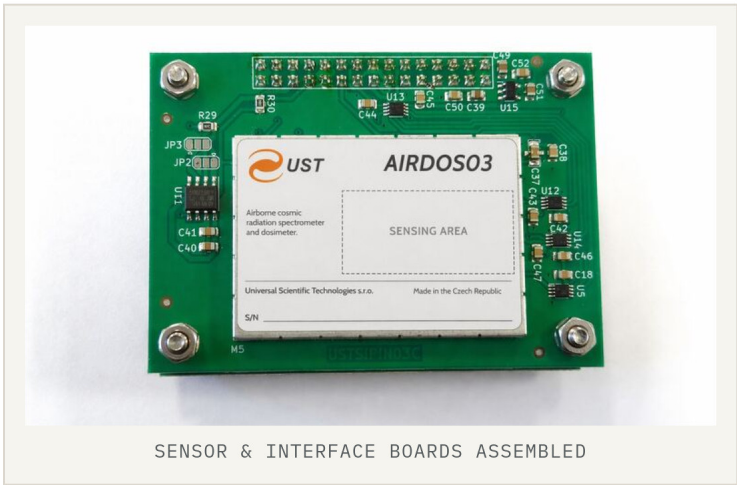
MAVLINK READY

PIXHAWK CONNECTOR

AIRDOS03

Lightweight Airborne Dosimeter & Spectrometer — UAVDOS

AIRDOS03 is a compact silicon PIN-diode ionizing radiation dosimeter and spectrometer for unmanned aerial vehicles. It measures the deposited energy of radiation events over a wide dynamic range and records temperature and humidity alongside, so radiation data can be corrected to local atmospheric conditions.



Flight heritage: integrated in the fuselage of an unmanned autogyro for in-situ measurements inside and near thunderstorms, and flown as part of the FIK-10 stratospheric balloon mission.

40 keV–80 MeV Deposited energy range	5 V · 3 mA Supply, typical	42 g Mass	71 × 51 mm Footprint, height 25 mm
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Applications

ATMOSPHERIC RESEARCH

- Unmanned atmospheric radiation surveys
- Detection of enhanced radiation in convective storm regions
- Mapping radiation gradients near ground or airborne sources

SPACE & HIGH-ALTITUDE RESEARCH

- Scientific support for space weather studies
- High-altitude dosimetry experiments
- Aviation radiation research

Main technical parameters

Deposited energy range	40 keV – 80 MeV
Energy resolution	15 ± 2 keV per channel (firmware-dependent)
Temperature drift	3.1 keV / °C
Effective energy channels	~65 000
Detection volume	44 mm ³
Spectrum integration time	10 s (configurable in firmware)
Temperature sensor	–40 to +125 °C, ±0.5 °C
Humidity sensor	0 to 100 %RH, ±2 %RH
Operating temperature	–40 to +50 °C
Operating humidity	20 to 80 %RH, non-condensing
Ingress protection	IP20
Supply voltage	4.5 – 5.4 V, typ. 3 mA
Data interface	6-pin UART, 115200 bps
Dimensions (W × H × D)	71 × 51 × 25 mm
Mass	42 g

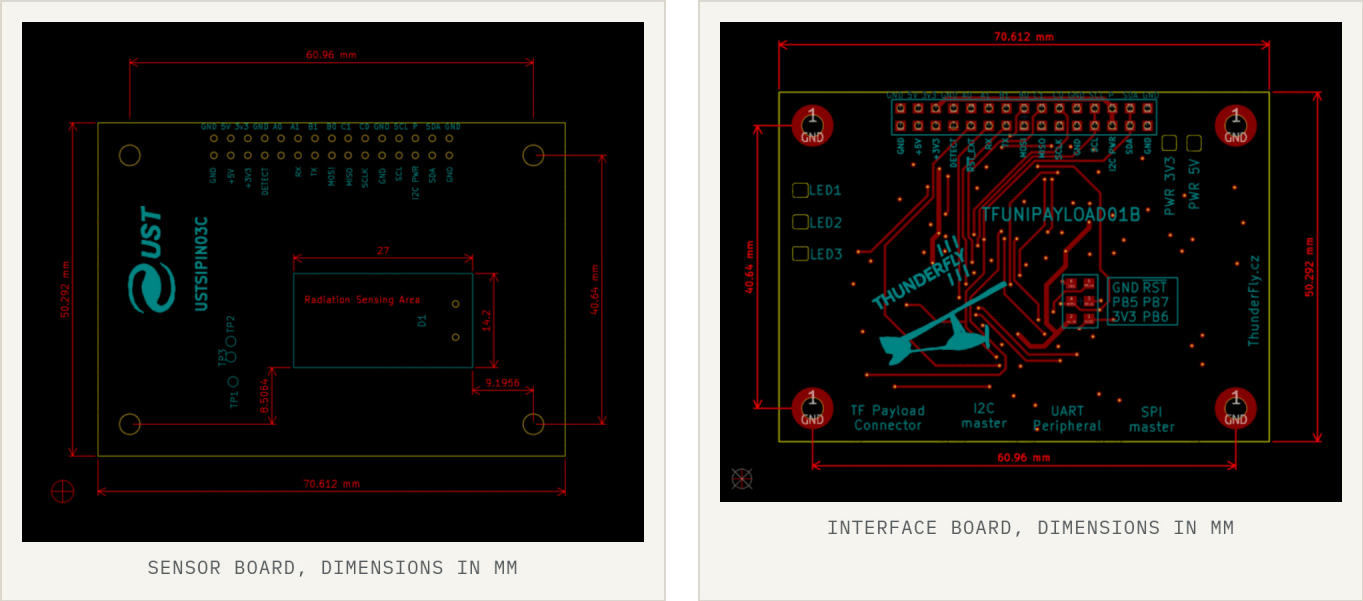
Meaning of measured data

AIRDOS03 measures the energy deposited in its silicon PIN diode by each ionizing radiation event. The energy-resolved spectrum is the primary product, integrated over a configurable period (10 s by default).

Dose-related quantities are derived from the deposited-energy spectrum over the detector volume and over time. Because radiation intensity at altitude depends on the local atmosphere, the integrated temperature and relative humidity readings are recorded with each measurement to support correction of the radiation data to local atmospheric conditions.

Synchronization with an external GNSS receiver (PPS time pulse on the auxiliary connector) allows each measurement to be time-tagged against the flight trajectory, attitude, and environmental telemetry recorded by the host platform.

Physical dimensions



Both boards measure 70.6 × 50.3 mm with 60.96 × 40.64 mm mounting holes. Boards are joined by a 2.54 mm pitch 15×2 pin header and stacked either in parallel (9 mm spacer columns) or in an L-shape with an angled header. Radiation sensing area: 27 × 14.2 mm.

Electrical interface

Two JST-GH connectors, Pixhawk-compatible.

UART (TELEM)	
PIN	FUNCTION
+5V	Power input, 5 V
RX	UART receive, to AIRDOS03
TX	UART transmit, from AIRDOS03
CTS	Clear to send (flow control)
RTS	Request to send (flow control)
GND	Ground

PAYLOAD (AUXILIARY)	
PIN	FUNCTION
TIMEPULSE	PPS / time pulse input
EXTINT	External interrupt input
GPIO	General-purpose I/O
SDA	I ² C data
SCL	I ² C clock
TX	Auxiliary UART transmit
RX	Auxiliary UART receive
GND	Ground

Data output

A continuous 115200 bps stream on the UART port. The firmware variant selects the format.

FIRMWARE	OUTPUT FORMAT	TYPICAL USE
MAVLink	MAVLink Tunnel packets	Integration with PX4 / ArduPilot autopilots, no autopilot firmware changes
UST DFF	UST Dosimeters File Format, ASCII	Stand-alone logging, USB or serial capture, SPACEDOS and AIRDOS toolchain

Python post-processing tools are provided with the MAVLink firmware: **mavlink_to_ardos.py** decodes a live or captured stream into xDOS text for the dosview log viewer; **ulog_to_ardos.py** extracts data from a PX4 ULog file; **flux_analysis.py** plots particle flux against time and altitude.

Firmware is open source and rebuilt with PlatformIO. Pre-built binaries are available in the project repository.

Ordering information: AIRDOS03 is available from Universal Scientific Technologies s.r.o. Special configurations and larger quantities, including integration support for airborne platforms, can be discussed on request. Contact sale@ust.cz for a commercial quotation.

Universal Scientific Technologies s.r.o.

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